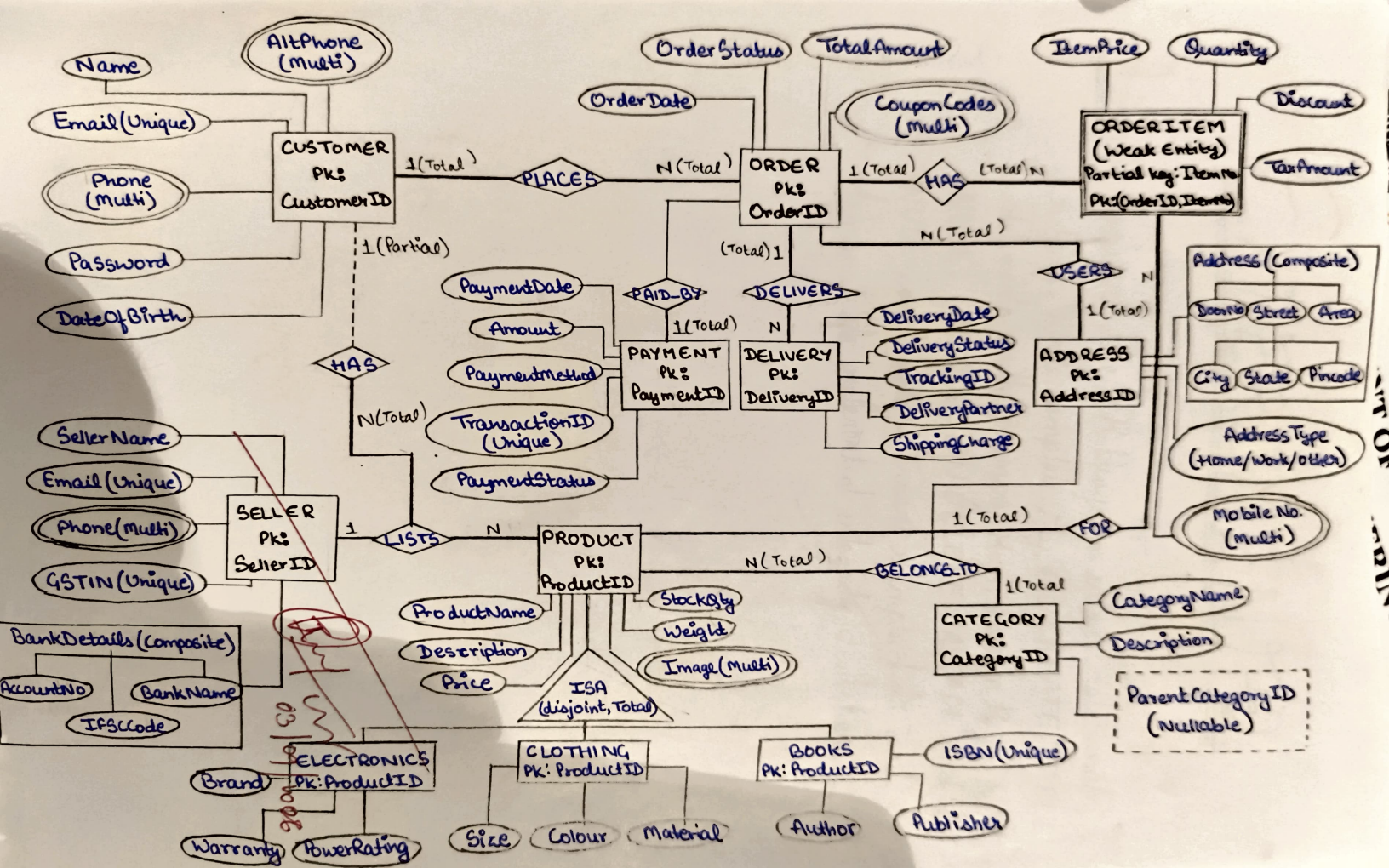
EXPERIMENT - 1

Aim :- Design an ER diagram for an Indian E-commerce platform with entities Customer, Product, Order, OrderItem, Seller, Category, Payment, Delivery, Address. Specify Primary keys, Composite and Multi-valued attributes, weak entities, specialization of Product and participation constraints.


Name
AltPhone (Multi)
Email (Unique)
Phone (Multi)
Password
DateOfBirth
CUSTOMER Pk: CustomerID
1 (Total)
PLACES
N (Total)
ORDER Pk: OrderID
OrderStatus
TotalAmount
OrderDate
CouponCodes (Multi)
1 (Total)
HAS
(Total) N
ORDERITEM (Weak Entity) Partial key: ItemNo Pk: (OrderID, ItemNo)
ItemPrice
Quantity
Discount
TaxAmount
1 (Partial)
HAS
N (Total)
(Total) 1
PAID-BY
DELIVERS
PaymentDate
Amount
PaymentMethod
TransactionID (Unique)
PaymentStatus
PAYMENT Pk: PaymentID
1 (Total)
N
DELIVERY Pk: DeliveryID
DeliveryDate
DeliveryStatus
TrackingID
DeliveryPartner
ShippingCharge
N (Total)
USERS
N
1 (Total)
ADDRESS Pk: AddressID
Address (Composite)
DoorNo
Street
Area
City
State
Pincode
AddressType (Home/Work/Other)
MobileNo. (Multi)
SellerName
Email (Unique)
Phone (Multi)
GSTIN (Unique)
SELLER Pk: SellerID
1
LISTS
N
PRODUCT Pk: ProductID
1 (Total)
FOR
N (Total)
BELONGS_TO
1 (Total)
CATEGORY Pk: CategoryID
CategoryName
Description
ParentCategoryID (Nullable)
BankDetails (Composite)
AccountNo
BankName
IFSCCode
ProductName
Description
Price
StockQty
Weight
Image (Multi)
ISA (disjoint, Total)
ELECTRONICS Pk: ProductID
Brand
Warranty
PowerRating
CLOTHING Pk: ProductID
Size
Colour
Material
BOOKS Pk: ProductID
ISBN (Unique)
Author
Publisher